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B.Tech. Degree VI Semester Examination April 2016

ME 1604 HEAT AND MASS TRANSFER

(2012 scheme)

(Use of approved data book is permitted)

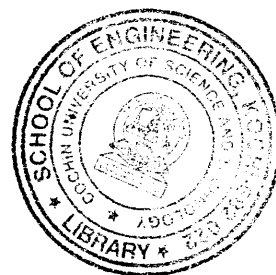
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Discuss on the various modes of heat transfer.
- (b) Define Biot number. What is its significance?
- (c) Explain Reynold's Analogy. What is its significance?
- (d) Differentiate drop wise condensation and film wise condensation.
- (e) Explain the terms absorptivity, reflectivity and transmissivity.
- (f) What is a perfectly black body? Differentiate a thermally black body and a visually black body.
- (g) What is meant by Fouling in heat exchangers? What is its effect on the performance of heat exchangers?
- (h) Compare the advantages and disadvantages of parallel flow and counter flow heat exchangers?



PART B

(4 × 15 = 60)

- II. (a) Derive Fourier's general heat conduction equation in Cartesian co-ordinate system. (7)
- (b) Circumferential aluminium fins ($k=200 \text{ W/mK}$) of rectangular profile (1.5 cm wide × 1.0 mm thick) are fitted on to a 2.5 cm diameter tube. The fin base temperature is 170°C and the ambient fluid temperature is 25°C . Estimate the heat loss per fin. The heat transfer coefficient ' h ' may be taken as $130 \text{ W/m}^2\text{K}$. (8)
- OR**
- III. A 1m long steel plate ($k = 50 \text{ W/m.K}$) is well insulated on its side while the top surface is at 100°C and the bottom surface is convectively cooled by a fluid at 20°C . Under steady state conditions with no heat generation, a thermocouple at the mid point reveals a temperature of 85°C . What is the value of the heat transfer coefficient at the bottom surface? (15)
- IV. Air at a pressure of 1 atm and at a temperature of 50°C is in parallel flow over the top surface of a flat plate that is heated to a uniform temperature of 100°C . The plate has a length of 20 cm and a width of 10 cm. The Reynolds number based on the plate length is 40,000. What is the rate of heat transfer from the plate to the air? (15)

OR

(P.T.O.)

- V. Engine oil is heated by flowing through a circular tube of diameter 50 mm and length 25 m and whose surface is maintained at 150°C . If the flow rate and inlet temperature of oil are 0.5 kg/s and 20°C , what is the outlet temperature of oil? What is the total heat transfer rate? (15)
- VI. (a) Explain the various laws of radiation. (5)
 (b) A pipe carrying steam, having an outside diameter of 20 cm runs in a large room and is exposed to air at a temperature of 30°C . The pipe surface temperature is 400°C . Calculate the loss of heat to surroundings per metre length of pipe due to thermal radiation. The emissivity of the pipe surface is 0.8. What would be the loss of heat due to radiation if the pipe is enclosed in a 40 cm diameter brick conduit of emissivity 0.91? (10)
- OR**
- VII. Two large parallel plates with emissivities 0.35 and 0.85 exchange heat by radiation. The plates are at 1073 K and 773 K. A radiation shield having the emissivity of 0.04 is placed between them. Find the percentage reduction in heat exchange and temperature of the shield. (15)
- VIII. (a) Derive an expression for the effectiveness of a parallel flow heat exchanger. (6)
 (b) Water enters a counterflow, double pipe heat exchanger at 15°C , flowing at the rate of 1300 kg/hr. It is heated by oil ($C_p = 2000 \text{ J/kgK}$) flowing at the rate of 550 kg/hr from the inlet temperature of 94°C . For an area of 1 m^2 and an overall heat transfer coefficient of $1075 \text{ W/m}^2\text{k}$, determine the total heat transfer and the outlet temperatures of water and oil. (9)
- OR**
- IX. A chemical having specific heat of 3.3 kJ/kg K flowing at the rate of 20000 kg/hr enters a parallel flow heat exchanger at 120°C . The flow rate of cooling water is 50000 kg/hr with an inlet temperature of 20°C . The heat transfer area is 10 m^2 and the overall heat transfer coefficient is $1050 \text{ W/m}^2\text{K}$. Find the effectiveness of the heat exchanger and the outlet temperature of water and chemical. (Take for water $C_p = 4.186 \text{ kJ/kg K}$). (15)

